

GENETIC SHIFT
A 'black' tiger (left) and its normal sibling at Nandankanan Zoological Park, Bhubaneswar

MAPPING THE BIG CAT

A team of researchers from Stanford, NCBS et al sequenced 65 genomes from four subspecies–Amur, Bengal, Sumatran and Malayan–over three years. The four were found to be genetically distinct. Indian tigers as a whole had the highest genetic variation, but some of them show signs of possible inbreeding. Even Bengal tigers, which comprise about 70 per cent of the world's wild tigers and exhibit relatively high genomic diversity compared to other subspecies, showed signs of inbreeding in some populations. Tigers from the Northeast were the most different from other populations in India. Extreme fragmentation and high human population density in India has resulted in isolated populations, where individuals may be more likely to mate with relatives. In contrast, despite low Amur tiger population densities in the Russian Far East, individual movement is not hindered by significant barriers and there is less inbreeding. Despite this, and other adaptations, Amur and Sumatran tigers have lower genetic diversity, so if populations continue to decline, genetic rescue may be needed.

for 28 months. The Rs 29 crore project that envisaged shifting three pairs of tigers to augment the dwindling pool at Satkosia was eventually dropped.

The researchers whose work forms the basis of the genetic rescue plan are from Stanford University, National Centre for Biological Sciences (NCBS) of the Tata Institute of Fundamental Research, Bengaluru, Wildlife Institute of India (WII), Dehradun, Swedish Museum of Natural History and several other reputed institutions. Their three papers buttressed the findings of a landmark 2014 Stanford study, published in the *Journal of Heredity*, which posited genetic diversity as the key to the survival of tigers in India. The latest findings say "fragmentation of habitat" has disrupted the natural evolutionary process in wild tigers and the situation will worsen as "anthropogenic pressures increase". Fragmented forests mean isolated, shrinking pools of subspecies, more inbreeding, lower reproduction rates, faster spread of disease, more cardiac defects...the stripes in a vicious loop.

NCBS professor Uma Ramakrishnan, co-author of one of the papers, says isolation has made the tigers of Ranthambore up to twice as inbred as those in larger, more connected populations of central and southern India. Officials say inbreeding depression has already resulted in a smaller litter size among Ranthambore tigers—mostly two, unlike the normal four or five. Tigers with "deleterious mutations" and reduced fitness are a genetic black hole—they bear less children, and are more vulnerable to disease. And, ultimately, to extinction. Besides translocation, the Stanford scholars recommend 'tiger corridors' that can enable mating across regions.

Not everyone is on board. Some experts prefer to let natural selection do its work unhindered—they believe weak mutants purge themselves via natural selection as the cubs carrying them are anyway prone to early death. "That takes care of inbreeding depression," says S.P. Yadav, NTCA member-secretary and additional director general, Project Tiger. Yadvendradev V. Jhala, WII dean and one of the co-authors, too, is among the anti-interventionists. "Genetic rescue should be seen as a last resort when populations manifest visible or physiological signs of inbreeding depression. There are no such signs as of now," he says. Research on neo-natal mortality due to mutant genes is limited, though, and not all bad mutations are purged. But the debate has shone light on a subtle truth—it's not a fight for mere numbers, but one against genetic poverty. ■